

# IDP/IAP FRAMEWORK TRANSFERABILITY

## GENERIC FRAMEWORK

### Detection Profile (DP)

$$DP = \langle \mathcal{I}, \mathcal{E}, \mathcal{S}, \mathcal{A} \rangle$$

*Identity • Evidence • Semantics • Analysis*

### Entity Profile (EP)

$$EP = \langle \mathcal{O}, \mathcal{G}, \mathcal{D} \rangle$$

*Observations • Geometry • Dependencies*

### Topology-aware Retrieval

$$\mathcal{N}_k(A_j)$$

*k-hop neighborhood expansion*

*Domain Instantiation*

### Building O&M [Validated]

**Entity:** BIM Element (Wall/Slab/Column)

**DP** → IDP (Integrated Defect Profile)

**EP** → IAP (Integrated Asset Profile)

**Topology:** Structural load paths

**Edge:** *supports, adjacent\_to*

### Renovation Planning [Demonstrated]

**Entity:** BIM Element + Age Metadata

**DP** → Condition Assessment Record

**EP** → Renovation Scope Profile

**Topology:** Renovation sequence

**Edge:** *precedes, depends\_on*

### Construction Progress [Conceptual]

**Entity:** Work Zone / Task Region

**DP** → WDP (Worker Detection Profile)

**EP** → WZP (Work Zone Profile)

**Topology:** Schedule dependencies

**Edge:** *successor\_of, blocked\_by*

*Invariant: EP structure ( $\mathcal{O}, \mathcal{G}, \mathcal{D}$ ) and k-hop retrieval remain constant across domains; only entity semantics and edge types vary.*